Section Overview

- Introduction to CUDA, SGEMM, and the Naive Kernel

What is GEMM?

Definition: Single-precision general matrix multiplication (SGEMM)

$$C \leftarrow \alpha AB + \beta C$$

What is GEMM?

Definition: Single-precision general matrix multiplication (SGEMM)

$$C \leftarrow \alpha AB + \beta C$$

- Setting $\alpha = 1$ and $\beta = 0$ gives ordinary matrix multiplication: $C = AB$.
- Other values let one call scale the product and accumulate it into an existing matrix.

What is GEMM?

Definition: Single-precision general matrix multiplication (SGEMM)

$$C \leftarrow \alpha AB + \beta C$$

- Setting $\alpha = 1$ and $\beta = 0$ gives ordinary matrix multiplication: $C = AB$.
- Other values let one call scale the product and accumulate it into an existing matrix.

Example

During the forward pass when training a model, we accumulate the gradients from the entire batch.

What is GEMM?

Definition: Single-precision general matrix multiplication (SGEMM)

$$C \leftarrow \alpha AB + \beta C$$

- Setting $\alpha = 1$ and $\beta = 0$ gives ordinary matrix multiplication: $C = AB$.
- Other values let one call scale the product and accumulate it into an existing matrix.

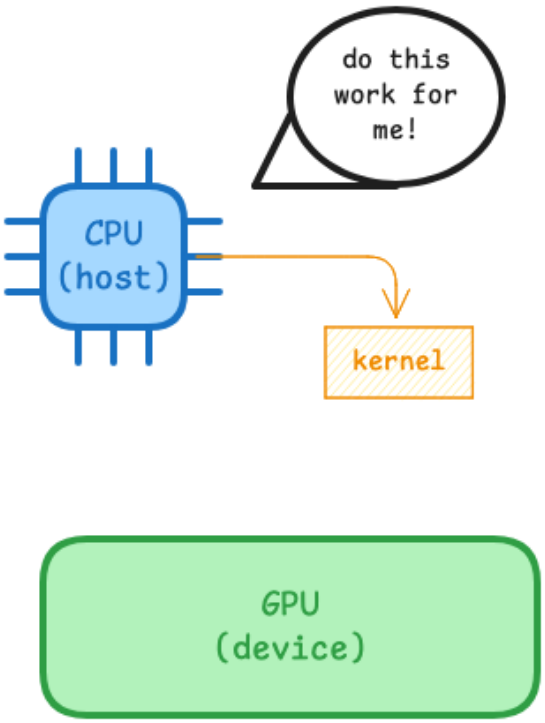
```
sgemm(
  int M, int N, int K,
  float alpha, float *A, float *B,
  float beta, float *C
)
```

$$\underbrace{A}_{M\times K} \quad \underbrace{B}_{K\times N} \quad \underbrace{C}_{M\times N}$$

- M: rows of A and C
- N: columns of B and C
- K: shared reduction dimension

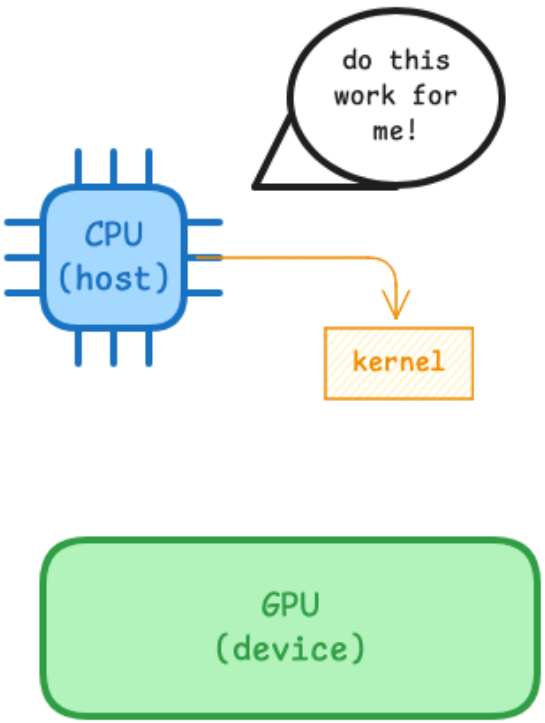
CUDA Basics (Host vs. Device)

- The CPU is known as the **host**, and manages top level logic.
- The GPU is known as the **device**, and does the heavy lifting.
- The host launches **kernels** to the device.



CUDA Basics (Host vs. Device)

- The CPU is known as the **host**, and manages top level logic.
- The GPU is known as the **device**, and does the heavy lifting.
- The host launches **kernels** to the device.




```
int main() {
  const int M = 128, N = 128, K = 128;

  float *dA, *dB, *dC;
  cudaMalloc(&dA, M * K * sizeof(float);
  cudaMalloc(&dB, K * N * sizeof(float));
  cudaMalloc(&dC, M * N * sizeof(float));

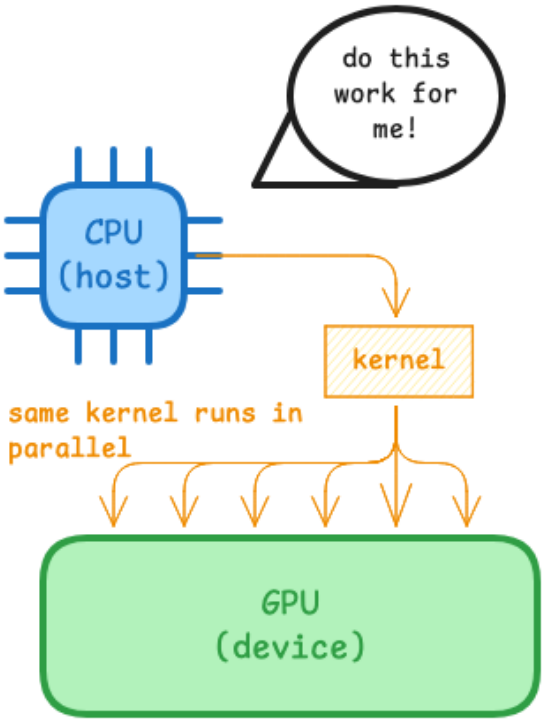
  // load values into these device pointers...

  run_sgemm_naive(M, N, K, 1, dA, dB, 0, dC);

  cudaFree(dA); cudaFree(dB); cudaFree(dC);
}
```

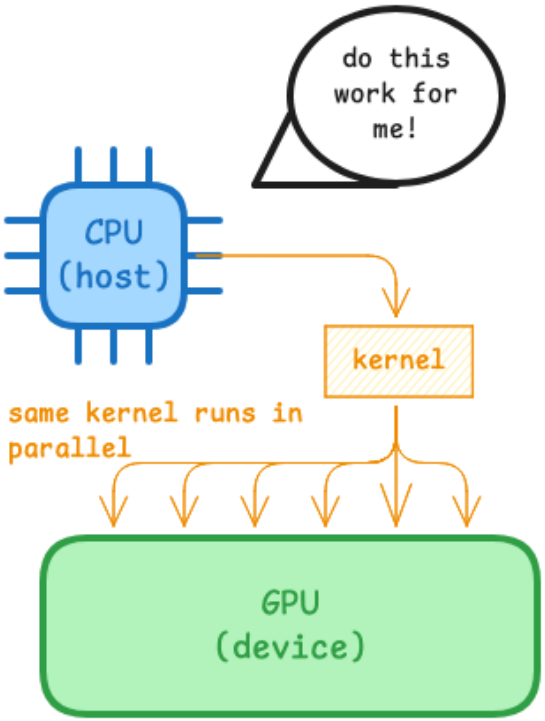
CUDA Basics (Kernel Execution)

- A kernel runs in parallel on many **threads** on the device.
- The number of threads the device allocates is configured by the host, when it **launches** the kernel.



CUDA Basics (Kernel Execution)

- A kernel runs in parallel on many **threads** on the device.
- The number of threads the device allocates is configured by the host, when it **launches** the kernel.

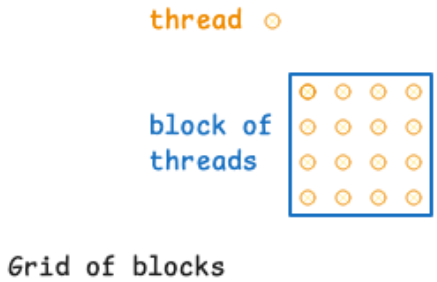



```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  // implementation hidden for now ...
}
```

```
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  constexpr int block_size = 32;
  const dim3 threads(block_size, block_size);
  const dim3 blocks((N + block_size - 1) / block_size,
                    (M + block_size - 1) / block_size);
  sgemm_naive<<<blocks, threads>>>(M, N, K, alpha, A, B, beta, C);
}
```

CUDA Basics (Threads, Blocks, Grids)

- Threads are grouped into **blocks**.
- The collection of blocks requested by the host is organized in a **grid**.




```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  // implementation hidden for now ...
}
```

```
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  constexpr int block_size = 32;
  const dim3 threads(block_size, block_size);
  const dim3 blocks((N + block_size - 1) / block_size,
                    (M + block_size - 1) / block_size);
  sgemm_naive<<<blocks, threads>>>(M, N, K, alpha, A, B, beta, C);
}
```

Why Threads, Blocks, Grids?

This seems complicated, why can't we just ask for N threads?

Why Threads, Blocks, Grids?

This seems complicated, why can't we just ask for N threads?

Relation to hardware

This hierarchy is closely related to how the device schedules the execution. We will see later how to use this to optimize performance.

Why Threads, Blocks, Grids?

This seems complicated, why can't we just ask for N threads?

Relation to hardware

This hierarchy is closely related to how the device schedules the execution.
We will see later how to use this to optimize performance.

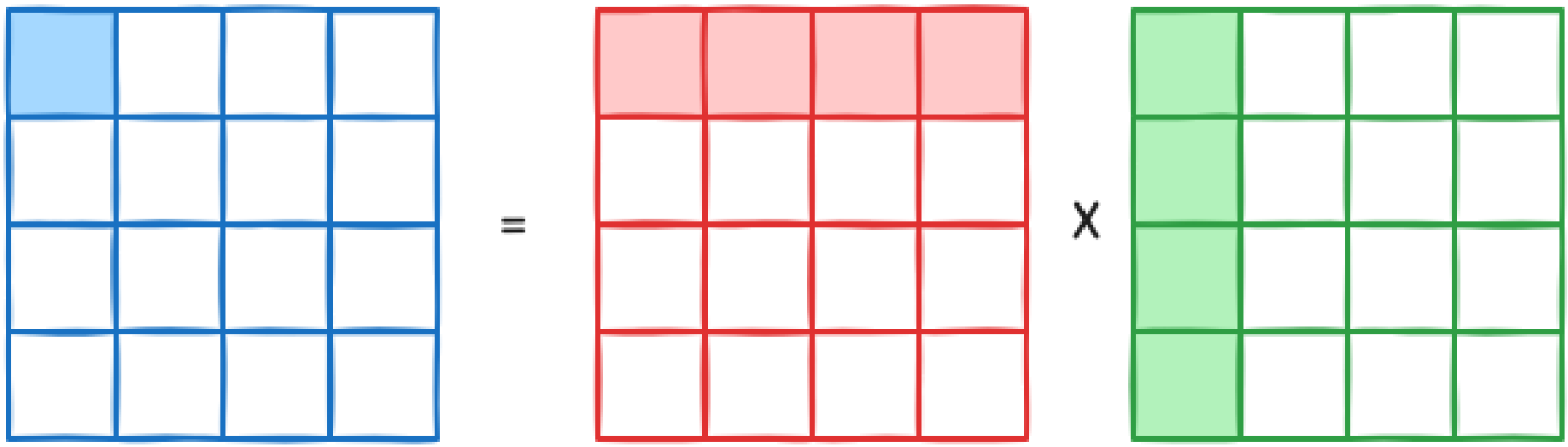
Thread, Block, Grid hierarchy

Threads are grouped into blocks of constant size/shape.
Blocks are launched as a grid.
We choose the dimensions of the grid to control how many blocks are launched, and thus how many threads are launched.

Let's design a SGEMM kernel!

Recall

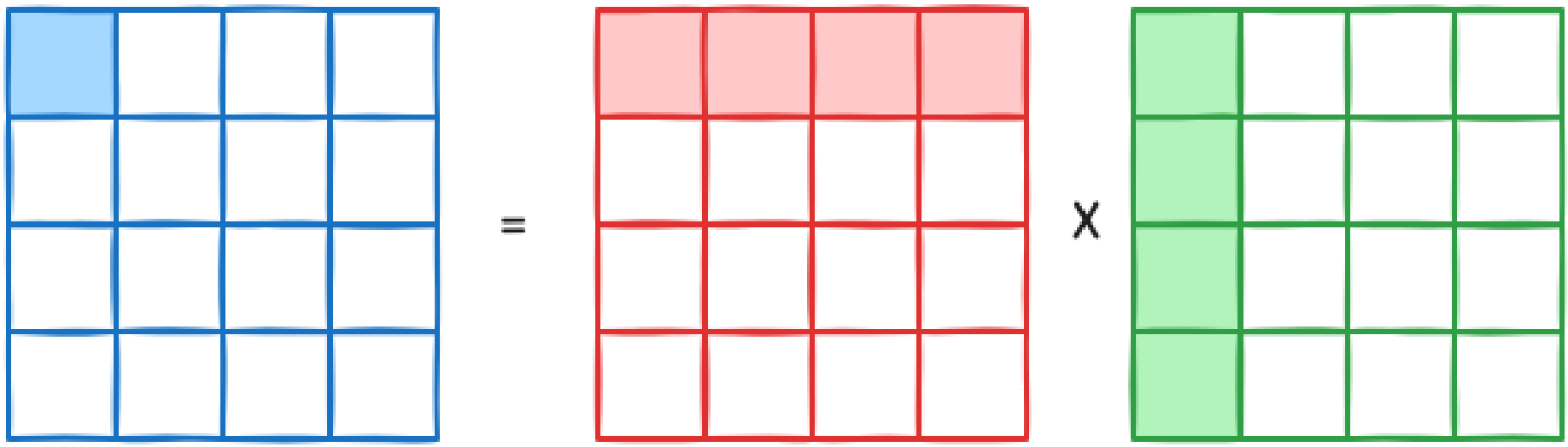
Recall that matrix multiplication is naturally parallel, and the output elements of C can be computed independently in parallel.



Let's design a SGEMM kernel!

Recall

Recall that matrix multiplication is naturally parallel, and the output elements of C can be computed independently in parallel.

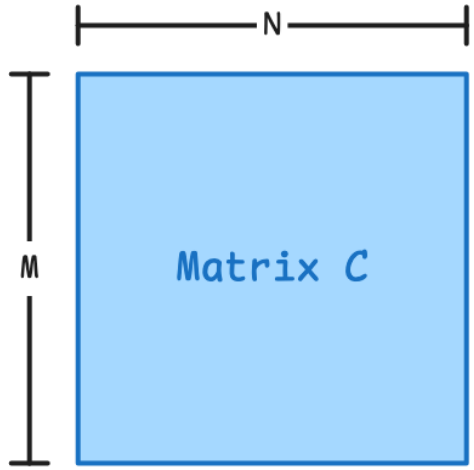


We can naturally map one thread to each output element!

Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

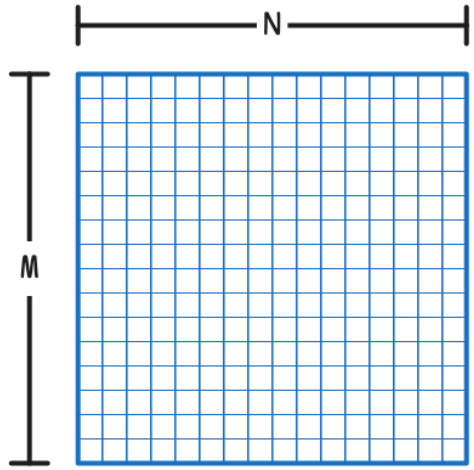
  if (y < M && x < N) {
    float sum = 0.0f;
    for (int i = 0; i < K; ++i) {
      sum += A[y * K + i] * B[i * N + x];
    }
    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  constexpr int block_size = 32;
  const dim3 threads(block_size, block_size);
  const dim3 blocks((N + block_size - 1) / block_size,
                    (M + block_size - 1) / block_size);
  sgemm_naive<<<blocks, threads>>>(M, N, K, alpha, A, B, beta, C);
}
```



Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

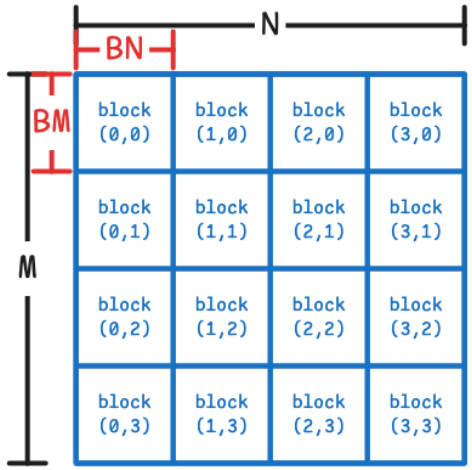
  if (y < M && x < N) {
    float sum = 0.0f;
    for (int i = 0; i < K; ++i) {
      sum += A[y * K + i] * B[i * N + x];
    }
    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  constexpr int block_size = 32;
  const dim3 threads(block_size, block_size);
  const dim3 blocks((N + block_size - 1) / block_size,
                    (M + block_size - 1) / block_size);
  sgemm_naive<<<blocks, threads>>>(M, N, K, alpha, A, B, beta, C);
}
```



Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

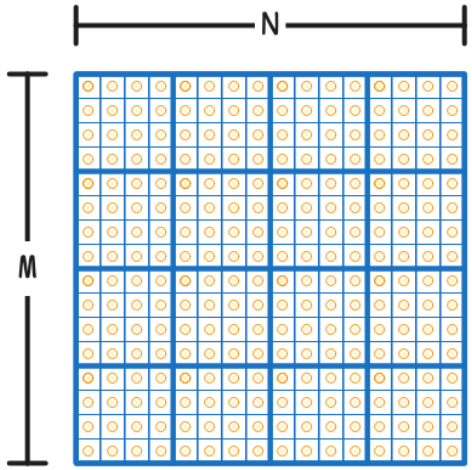
  if (y < M && x < N) {
    float sum = 0.0f;
    for (int i = 0; i < K; ++i) {
      sum += A[y * K + i] * B[i * N + x];
    }
    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  constexpr int block_size = 32;
  const dim3 threads(block_size, block_size);
  const dim3 blocks((N + block_size - 1) / block_size,
                    (M + block_size - 1) / block_size);
  sgemm_naive<<<blocks, threads>>>(M, N, K, alpha, A, B, beta, C);
}
```



Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

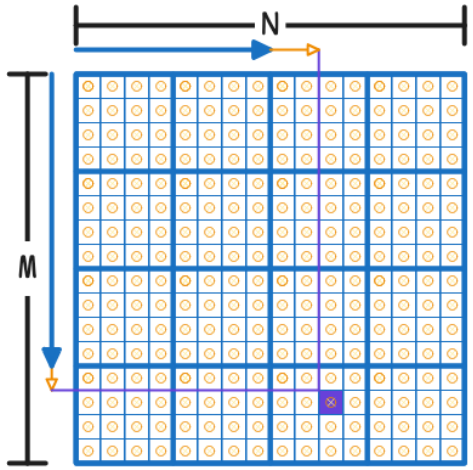
  if (y < M && x < N) {
    float sum = 0.0f;
    for (int i = 0; i < K; ++i) {
      sum += A[y * K + i] * B[i * N + x];
    }
    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  constexpr int block_size = 32;
  const dim3 threads(block_size, block_size);
  const dim3 blocks((N + block_size - 1) / block_size,
                    (M + block_size - 1) / block_size);
  sgemm_naive<<<blocks, threads>>>(M, N, K, alpha, A, B, beta, C);
}
```



Naive SGEMM Kernel

```
__global__ void sgemm_naive(const int M, const int N, const int K,
                            const float alpha, const float *A, const float *B,
                            const float beta, float *C) {
  const int y = blockIdx.y * blockDim.y + threadIdx.y;
  const int x = blockIdx.x * blockDim.x + threadIdx.x;

  if (y < M && x < N) {
    float sum = 0.0f;
    for (int i = 0; i < K; ++i) {
      sum += A[y * K + i] * B[i * N + x];
    }
    C[y * N + x] = alpha * sum + beta * C[y * N + x];
  }
}
void launch_sgemm_naive(const int M, const int N, const int K,
                        const float alpha, const float *A, const float *B,
                        const float beta, float *C) {
  constexpr int block_size = 32;
  const dim3 threads(block_size, block_size);
  const dim3 blocks((N + block_size - 1) / block_size,
                    (M + block_size - 1) / block_size);
  sgemm_naive<<<blocks, threads>>>(M, N, K, alpha, A, B, beta, C);
}
```



Naive SGEMM visualization

- One thread computes one element of C.
- It walks one row of A and one column of B for the full dot product.

Play naive-kernel visualization

Naive CUDA is already faster than the CPU

